BIO 004 Human Anatomy · Module 4 notes · Dr. Sharilyn Rennie

The digestive system, part II: stomach, pancreas, liver, and gallbladder

The written notes for the second digestive lecture: the stomach, the pancreas, the liver and its lobule, the gallbladder and the path of bile, and the accessory organ disorders. Module 4 competencies 115 and 122 to 125, with nine worked questions.

To print: these notes as a two-column PDF (opens in a new tab). Already printed the Module 4 notes packet? These are the lecture notes, a different document, so print these too if you want them on paper.

How to use these notes. Draw the note sheet for each competency first, from memory, before you read. Then read the matching section here to fill in anything you missed, and use it to review. Every section carries its competency number, so you can go from the competency packet to the page and back. The worked questions at the end show how a lecture exam asks about this material and why each wrong answer is wrong.

Competency 115. Stomach regions and features

In your course notes: Module 4 packet, Chapter 3, competency 115 (page 32).

Dr. Rennie's video for this part: Digestive system concept video, chapter Stomach, pancreas, liver, gallbladder.

You should be able to: Identify the cardia, fundus, body, pyloric part, curvatures, rugae, and the three-layered muscularis of the stomach.

The stomach is a J-shaped widening of the canal just under the diaphragm, connecting the esophagus above to the duodenum below. It is a mixing chamber and a holding tank: it stores the meal, churns it with gastric juice until the semisolid bolus is a liquid called chyme, begins the digestion of starch, triglycerides, and proteins, absorbs a few substances, and releases chyme into the duodenum a little at a time. Empty, it is about the size of a large sausage. Full, it stretches to hold 2 to 3 liters.

The four regions

Region	Where it is
Cardia (cardiac region)	The small region around the opening from the esophagus
Fundus	The rounded dome that bulges above and to the left of the cardia, up under the diaphragm
Body	The large central part below the fundus
Pyloric part (pyloric region)	The funnel that connects the body to the duodenum, in three parts in order: the pyloric antrum (wide, joins the body), the pyloric canal (narrow), and the pylorus (the outlet, surrounded by the pyloric sphincter)

Lesser curvature

The short, concave border on the right (medial) side, facing the liver. The lesser omentum attaches here and runs up to the liver.

Greater curvature

The long, convex border on the left (lateral) side. The greater omentum hangs from here and drapes over the intestines.

Rugae
Large folds of mucosa and submucosa on the inside of an empty stomach. They flatten as the stomach fills, which is how it stretches.

Pyloric sphincter
The ring of smooth muscle around the pylorus that meters chyme into the duodenum. A thickening of the circular layer.

- Draw the fundus up beside the esophagus, not at the bottom of the J. Food collects there, but it does not have to pass through it to move on.
- Our notes call the curvatures medial and lateral (OpenStax); your slides call them superior and inferior. Both describe the same two borders.

The wall of the stomach

Layer	In the stomach
Mucosa	Simple columnar epithelium of surface mucous cells. The surface is dotted with gastric pits, narrow channels that gastric glands open into. Lamina propria of areolar connective tissue; muscularis mucosae of smooth muscle
Submucosa	Areolar connective tissue
Muscularis	Three layers of smooth muscle instead of two: inner oblique (the extra layer, mostly in the body), middle circular (thickens into the pyloric sphincter), outer longitudinal. The oblique fibers let the stomach twist and squeeze as well as push
Serosa	Visceral peritoneum. At the lesser curvature it continues up to the liver as the lesser omentum; at the greater curvature it continues down as the greater omentum

The cells of the gastric glands

Cell	Secretes
Mucous neck cells	Mucus
Chief cells	Pepsinogen (the inactive form of the protein-digesting enzyme pepsin) and gastric lipase
Parietal cells	Hydrochloric acid, and intrinsic factor, which is needed to absorb vitamin B12
G cells (enteroendocrine), mostly in the pyloric antrum	Gastrin, a hormone that stimulates gastric activity

Together the exocrine secretions are gastric juice.

What this means for patients

- Pylorospasm: in some infants the pyloric sphincter fails to relax, food cannot leave the stomach, the stomach overfills, and the baby vomits to relieve the pressure. Treated with drugs that relax the sphincter.
- Pyloric stenosis: a narrowing of the pyloric sphincter in infants. The hallmark is projectile vomiting. It needs surgical correction.
- Vomiting (emesis) is the forceful expulsion of the contents of the stomach, and sometimes the duodenum, through the mouth. Irritation or stretch of the stomach, unpleasant sights, general anesthesia, dizziness, and drugs such as morphine and digitalis all signal the vomiting center in the medulla (the area postrema). The diaphragm and abdominal muscles squeeze the stomach and the contents come up through relaxed esophageal sphincters. Prolonged vomiting in infants and the elderly is dangerous: loss of acidic gastric juice shifts blood pH toward alkalosis, the person dehydrates, and acid damages the esophagus and teeth.

Competency 124. Pancreas structure

In your course notes: Module 4 packet, Chapter 4, competency 124 (page 44).

Dr. Rennie's video for this part: Digestive system concept video, chapter Stomach, pancreas, liver, gallbladder.

You should be able to: Identify the head, body, and tail of the pancreas with its main and accessory ducts, major duodenal papilla, sphincter of Oddi, acini, and islets.

Most chemical digestion happens in the small intestine, and the pancreas, liver, and gallbladder supply the tools.

The pancreas is a long, soft gland lying across the posterior abdominal wall behind the greater curvature of the stomach. It is retroperitoneal. It is an exocrine gland (pancreatic juice down ducts into the duodenum) and an endocrine gland (hormones into the blood).

Structure	What it is
Head	The wide end, in the C-shaped curve of the duodenum, right of the midline
Body	The middle part, crossing the midline behind the stomach, in front of the aorta and the vertebral column
Tail	The narrow left end, near the spleen
Main pancreatic duct (duct of Wirsung)	The larger duct. Runs the whole length of the gland from tail to head, collecting pancreatic juice. In the wall of the duodenum it joins the common bile duct at the hepatopancreatic ampulla (ampulla of Vater)
Accessory pancreatic duct (duct of Santorini)	The smaller duct. Drains part of the head and opens on its own into the duodenum at the minor duodenal papilla, a little above the major papilla
Hepatopancreatic ampulla	The short widened chamber where the common bile duct and main pancreatic duct meet
Major duodenal papilla	The bump in the duodenal lining where the ampulla opens, in the descending part of the duodenum below the pyloric sphincter
Sphincter of Oddi (hepatopancreatic sphincter)	The ring of smooth muscle around the ampulla that controls bile and pancreatic juice flowing into the duodenum
Acini	Small clusters of gland cells around tiny ducts: the exocrine part, about 99 percent of the gland. They make pancreatic juice, 1200 to 1500 mL a day of clear fluid: water, salts, sodium bicarbonate, and digestive enzymes
Pancreatic islets (islets of Langerhans)	Pale round patches scattered among the acini: the endocrine part, about 1 percent. They release glucagon, insulin, somatostatin, and pancreatic polypeptide into the blood

- Match the size words. The main duct opens at the major papilla. The accessory duct opens at the minor papilla.
- Only the main duct joins the common bile duct. The accessory duct empties on its own.

Competency 122. Liver gross and lobule anatomy

In your course notes: Module 4 packet, Chapter 4, competency 122 (page 40).

Dr. Rennie's video for this part: Digestive system concept video, chapter Stomach, pancreas, liver, gallbladder.

You should be able to: Identify the four liver lobes, falciform ligament, and ligamentum teres, and name the hepatocytes, sinusoids, canaliculi, and portal triad of the lobule.

The liver is the largest gland in the body. It sits under the diaphragm, filling most of the right hypochondriac region and part of the epigastric region, protected by the lower ribs. Visceral peritoneum covers most of it, and under that a capsule of dense irregular connective tissue covers all of it.

Lobes and ligaments

Structure	Where and what
Right lobe	The largest lobe, right of the falciform ligament
Left lobe	Left of the falciform ligament, reaching over the stomach
Quadrate lobe	A small square lobe on the inferior surface, between the gallbladder and the ligamentum teres
Caudate lobe	A small lobe on the posterior surface beside the inferior vena cava
Porta hepatis	The doorway between the quadrate and caudate lobes. The hepatic portal vein and hepatic artery proper enter here; the right and left hepatic ducts leave here
Falciform ligament	The sickle-shaped fold of peritoneum that suspends the liver from the diaphragm and the anterior abdominal wall and marks the line between the right and left lobes on the front
Ligamentum teres (round ligament)	The fibrous cord in the free edge of the falciform ligament, running from the umbilicus to the liver. It is what remains of the fetal umbilical vein
Coronary ligaments	Narrow extensions of parietal peritoneum that hang the liver from the diaphragm
Lesser omentum	The fold from the lesser curvature of the stomach and the start of the duodenum to the visceral surface of the liver. Its free edge carries the portal vein, hepatic artery proper, and common bile duct

- From the front you see only two lobes. The quadrate and caudate lobes are underneath and behind.
- The ligamentum teres comes from the umbilical vein, which carried oxygen-rich blood from the placenta. Vein, not artery.

The liver lobule

Liver tissue is thousands of small six-sided units. Hepatocytes are the working cells: specialized epithelial cells with 5 to 12 sides making up about 80 percent of the liver's volume. They are arranged in plates one cell thick, the hepatic laminae, that radiate out from a central vein like spokes. Between the plates run the sinusoids.

Structure	Where it is	What it does
Central vein	The center of the lobule	Collects blood from the sinusoids
Hepatocytes	In plates (hepatic laminae) radiating from the central vein	The working cells: they make bile and do the liver's metabolic work
Hepatic sinusoids	Between the plates	Wide, leaky, endothelium-lined capillaries carrying mixed blood from the edge of the lobule toward the central vein. Stellate macrophages (Kupffer cells) line them and destroy worn-out blood cells, bacteria, and foreign matter arriving from the gut
Bile canaliculi	Grooves between neighboring hepatocytes inside each plate	Collect the bile the hepatocytes make and carry it outward to the edge of the lobule
Portal triad	At each corner of the lobule	A branch of the hepatic portal vein, a branch of the hepatic artery, and a bile duct. The two vessels bring blood in; the duct carries bile out

- Blood flows in toward the central vein. Bile flows out toward the portal triad. Opposite directions.
- Bile is a yellow-brown, sometimes olive-green liquid. It is both an excretory product and a digestive secretion.

Blood through the liver, in order

1. The hepatic artery proper (oxygen-rich blood from the aorta) and the hepatic portal vein (oxygen-poor blood carrying newly absorbed nutrients, drugs, microbes, and toxins from the gut and spleen) enter at the porta hepatis.

2. Branches of both run to the portal triads at the corners of every lobule.
3. The two streams mix in the sinusoids and flow past the hepatocytes toward the center.
4. The central vein collects the blood.
5. Central veins join into hepatic veins, which leave the back of the liver.
6. The hepatic veins empty into the inferior vena cava just under the diaphragm, and the blood returns to the right atrium.

Because so much blood from the gut passes through the liver, it is a very common site for cancers of the digestive tract to spread.

What the liver does

- Makes bile and the bile salts in it, which emulsify fats in the small intestine so they can be absorbed.
- Carbohydrate metabolism: keeps blood glucose steady and stores extra glucose as glycogen.
- Lipid metabolism and protein metabolism, including removing the amino group from amino acids (deamination) and sending the ammonia out as urea in the urine.
- Processes drugs and hormones, including alcohol and antibiotics.
- Excretes bilirubin, the pigment left over from breaking down old red blood cells.
- Stores glycogen and vitamins A, B12, D, E, and K.
- Phagocytosis by the Kupffer cells, and the final activation of vitamin D.

Competency 123. Gallbladder and bile pathway

In your course notes: Module 4 packet, Chapter 4, competency 123 (page 42).

Dr. Rennie's video for this part: Digestive system concept video, chapter Stomach, pancreas, liver, gallbladder.

You should be able to: Identify the gallbladder regions and trace bile from the canaliculi through the hepatic, cystic, and common bile ducts to the hepatopancreatic ampulla and duodenum.

The gallbladder is a pear-shaped muscular sac about 8 to 10 cm long, sitting in a shallow hollow on the visceral surface of the right lobe of the liver and hanging a little past its lower edge. It stores bile between meals and concentrates it by removing water. When fatty food reaches the duodenum, the smooth muscle in its wall contracts and squeezes bile out.

Structure	What it is
Fundus	The wide, rounded end that sticks out past the lower border of the liver
Body	The main middle part
Neck	The narrow end that points toward the porta hepatis and continues as the cystic duct
Right and left hepatic ducts	Drain bile from the two sides of the liver and leave at the porta hepatis
Common hepatic duct	Formed where the right and left hepatic ducts join
Cystic duct	Runs from the neck of the gallbladder and joins the common hepatic duct. Carries bile in both directions
Common bile duct	Formed where the common hepatic duct and the cystic duct join. Runs down in the edge of the lesser omentum, behind the first part of the duodenum, and through the head of the pancreas
Hepatopancreatic ampulla, sphincter of Oddi, major duodenal papilla	Where the common bile duct meets the main pancreatic duct and empties into the descending duodenum (competency 124)

A drop of bile, from the liver to the duodenum

1. Hepatocytes make the bile.
2. Bile canaliculi between the hepatocytes collect it and carry it to the edge of the lobule.

3. Bile ductules and the bile ducts of the portal triads carry it out of the lobules and join into larger ducts.
4. Right and left hepatic ducts leave the liver at the porta hepatis.
5. They join to form the common hepatic duct.
6. Side trip: when the sphincter of Oddi is closed, bile backs up and turns up the cystic duct into the gallbladder for storage. When the gallbladder contracts, bile comes back down the same cystic duct.
7. Common hepatic duct plus cystic duct form the common bile duct.
8. At the hepatopancreatic ampulla the common bile duct joins the main pancreatic duct.
9. The sphincter of Oddi opens and bile enters the duodenum through the major duodenal papilla, where it mixes with food.

- The duct names tell you where each duct comes from. Hepatic means liver. Cystic means bladder, this bladder. The common bile duct is where both come together.
- The cystic duct joins the common hepatic duct. It does not run straight to the duodenum. The gallbladder is a side trip, not a step every drop takes.

Competency 125. Clinical anatomy of accessory disorders

In your course notes: Module 4 packet, Chapter 4, competency 125 (page 45).

You should be able to: Match gallstones, cholecystitis, hepatitis, cirrhosis, jaundice, pancreatitis, and mumps to the accessory organ each one affects and explain how a stone at the ampulla can block bile and pancreatic flow at once.

Disorder	Organ affected	What changes
Gallstones	Gallbladder and bile ducts	Hard deposits form from bile in the gallbladder. A stone can move into the cystic duct or the common bile duct and block it
Cholecystitis	Gallbladder	The wall is inflamed and swollen, most often because a stone blocks the cystic duct and traps bile inside
Hepatitis	Liver	The liver is inflamed and hepatocytes are damaged, most often by a virus, also by alcohol or some drugs
Cirrhosis	Liver	After long-term damage, hepatocytes are replaced by bands of scar tissue and lumps of regrowing cells. The lobules lose their pattern and blood has trouble getting through
Jaundice	Liver or bile ducts, seen in the skin and eyes	Bilirubin builds up in the blood and stains the skin and the whites of the eyes yellow, because the liver cannot clear it into bile, a duct is blocked, or red cells break down too fast
Pancreatitis	Pancreas	The pancreas is inflamed and swollen. Digestive enzymes become active inside the gland and digest its own tissue. Gallstones and alcohol are common causes
Mumps	Parotid glands	A viral infection swells the parotids, so the face puffs out in front of and below the ears

Shared ducts, shared problems

- The common bile duct and the main pancreatic duct meet at the hepatopancreatic ampulla and share one exit into the duodenum. A gallstone that travels down the common bile duct and lodges in the ampulla plugs that shared exit.
- Bile backs up into the liver and the blood: jaundice.
- Pancreatic juice backs up into the pancreas, where its enzymes damage the gland: pancreatitis.
- One stone, two organs. That is one way gallstones lead to pancreatitis.
- Mumps swells the parotid gland, not the pancreas. Cholecystitis is the gallbladder; hepatitis and cirrhosis are the liver.

Worked questions

Nine questions in the style of the lecture exam. Read the answer and the reasons each wrong option is wrong. The wrong options are where the learning is.

QUESTION 1 · COMPETENCY 115

The dome-shaped region of the stomach that bulges above and to the left of the esophageal opening is the

A. cardia

B. fundus

C. body

D. pyloric antrum

Answer: B. The fundus is the rounded dome up under the diaphragm, above and left of the cardia. Food can collect there without passing through it.

Why the others are wrong

- **A** The cardia is the small region right around the opening from the esophagus.
- **C** The body is the large central part below the fundus.
- **D** The pyloric antrum is the wide first part of the funnel at the exit of the stomach.

QUESTION 2 · COMPETENCY 115

Which muscle layer is found in the stomach but not in the rest of the canal?

A. Inner oblique

B. Middle circular

C. Outer longitudinal

D. Muscularis mucosae

Answer: A. The stomach has three layers of smooth muscle. The inner oblique layer is the extra one; its diagonal fibers let the stomach twist and squeeze food.

Why the others are wrong

- **B** The circular layer is in the whole tube. In the stomach it is the middle layer and thickens at the pylorus.
- **C** The longitudinal layer is in the whole tube as the outer layer.
- **D** The muscularis mucosae is the thin muscle layer inside the mucosa of the whole tube, not a layer of the muscularis.

QUESTION 3 · COMPETENCY 115

Which gastric gland cell secretes hydrochloric acid and intrinsic factor?

A. Chief cell

B. Mucous neck cell

C. Parietal cell

D. G cell

Answer: C. Parietal cells make hydrochloric acid and intrinsic factor, the protein needed to absorb vitamin B12.

Why the others are wrong

- **A** Chief cells secrete pepsinogen and gastric lipase.
- **B** Mucous neck cells secrete mucus.
- **D** G cells are enteroendocrine cells that release the hormone gastrin into the blood.

QUESTION 4 · COMPETENCY 124

Which pancreatic duct joins the common bile duct at the hepatopancreatic ampulla?

A. The accessory duct (duct of Santorini)

B. The main pancreatic duct (duct of Wirsung)

C. Both ducts

D. Neither; pancreatic juice enters at the minor papilla only

Answer: B. The main pancreatic duct runs the whole length of the gland and joins the common bile duct at the ampulla, which opens at the major duodenal papilla.

Why the others are wrong

- **A** The accessory duct drains part of the head and opens on its own at the minor duodenal papilla.
- **C** Only the main duct joins the bile duct. The accessory duct empties separately.
- **D** Most pancreatic juice enters through the ampulla at the major papilla. Only the accessory duct uses the minor papilla.

QUESTION 5 · COMPETENCY 124

On a slide of the pancreas, the pale round patches among the darker clusters are

A. acini, the exocrine part

B. pancreatic islets, the endocrine part

C. bile canaliculi

D. hepatic sinusoids

Answer: B. The islets of Langerhans are the pale round patches, about 1 percent of the gland, releasing hormones such as insulin and glucagon into the blood.

Why the others are wrong

- **A** The acini are the darker clusters around them, about 99 percent of the gland, making pancreatic juice.
- **C** Bile canaliculi are in the liver, between hepatocytes.
- **D** Sinusoids are the leaky capillaries of the liver lobule, not pancreatic tissue.

QUESTION 6 · COMPETENCY 122

The ligamentum teres is the remnant of which fetal vessel?

A. Umbilical artery

B. Umbilical vein

C. Ductus arteriosus

D. Ductus venosus

Answer: B. The ligamentum teres (round ligament) runs from the umbilicus to the liver in the free edge of the falciform ligament. Before birth it was the umbilical vein carrying oxygen-rich blood from the placenta.

Why the others are wrong

- **A** The umbilical arteries become the medial umbilical ligaments on the inside of the anterior abdominal wall.
- **C** The ductus arteriosus was the fetal shunt between the pulmonary trunk and the aorta; it becomes the ligamentum arteriosum.
- **D** The ductus venosus was the fetal shunt through the liver; it becomes the ligamentum venosum, not the round ligament.

QUESTION 7 · COMPETENCY 122

In a liver lobule, blood flows

A. from the central vein outward to the portal triads

B. from the portal triads through the sinusoids to the central vein

C. from the bile canaliculi into the sinusoids

D. from the hepatic veins into the portal triads

Answer: B. The two vessels of each portal triad feed the sinusoids, and the blood flows past the hepatocytes toward the central vein in the middle. Bile goes the opposite way, out toward the triads.

Why the others are wrong

- **A** That is the direction of bile, not blood.
- **C** Bile canaliculi carry bile and do not connect to the sinusoids.
- **D** Hepatic veins carry blood out of the liver to the inferior vena cava. They are downstream of the central veins, not upstream of the triads.

QUESTION 8 · COMPETENCY 123

The common bile duct is formed by the union of the

A. right and left hepatic ducts

B. common hepatic duct and the cystic duct

C. cystic duct and the main pancreatic duct

D. common hepatic duct and the accessory pancreatic duct

Answer: B. The right and left hepatic ducts first form the common hepatic duct. The cystic duct from the gallbladder joins it, and from that point the duct is the common bile duct.

Why the others are wrong

- **A** The right and left hepatic ducts form the common hepatic duct, one step earlier.
- **C** The main pancreatic duct joins the common bile duct further down, at the hepatopancreatic ampulla. The cystic duct never meets it directly.
- **D** The accessory pancreatic duct opens on its own at the minor papilla and joins nothing.

QUESTION 9 · COMPETENCY 125

A gallstone lodges in the hepatopancreatic ampulla. Which two problems can follow?

A. Jaundice and pancreatitis

B. Mumps and hepatitis

C. Cholecystitis and cirrhosis

D. Hepatitis and pancreatitis

Answer: A. The ampulla is the shared exit for the common bile duct and the main pancreatic duct. Blocked, bile backs up into the liver and blood (jaundice) and pancreatic enzymes back up into the pancreas (pancreatitis).

Why the others are wrong

- **B** Mumps is a viral infection of the parotid glands; it has nothing to do with a stone. Hepatitis is inflammation of the liver by a virus, alcohol, or drugs.
- **C** Cholecystitis usually follows a stone in the cystic duct, which traps bile in the gallbladder, not a stone at the ampulla. Cirrhosis is the scarring left by long-term liver damage.
- **D** Pancreatitis is right, but hepatitis is inflammation of the liver by a virus or toxin, not a blocked duct. The blocked duct produces jaundice.